A Minimax Barrier for Neural Scaling Laws, and a Sharp Criterion for When Emergence Is Real

Ivan Drozdov

research@drozdov.space | code & benchmark: github.com/drozdisme/spectral_scaling

The question. The Chinchilla loss $L(N, D) = c_1 N^{-\alpha} + c_2 D^{-\beta} + L_0$ is fit to data, not derived from first principles. The debate over emergent abilities splits between Schaeffer's "mirage" and Wei's "real"—with no mathematical object to decide which reading applies to a given task. This paper supplies that object and, as a consequence, explains both the exponents and the phase transitions. The full programme spans six manuscripts; this note is the two-page map.

1. The setup and the three open questions

Three questions sit unanswered behind modern deep learning. **(Q1)** What pins the exponents α and β? **(Q2)** Are capability jumps real discontinuities or metric artifacts? **(Q3)** Which structures let knowledge transfer—the fourth theory of information, missing alongside computability, complexity, and learnability.

What triggered the investigation: running kernel ridge regression on a planted Sobolev spectrum, the source exponent r refused to exceed $\frac{1}{2}$ regardless of initialisation. The ceiling matched a minimax bound I had not connected to feature learning. Everything below is the explanation of why that ceiling is exact and universal.

The setting that makes all three tractable at once: data on a compact Riemannian manifold $\mathcal{M}$ of intrinsic dimension d^*, with Sobolev-s kernel eigenvalues $\sigma_k \asymp k^{-2s/d^*}$ (Weyl). A target f^* has spectral coefficients $\bar{a}_k^2 \asymp k^{-(1+2t)}$; parameter $t > 0$ is the target's *source exponent* relative to the Sobolev kernel, left free rather than fixed to the border of $\mathcal{H}^s$. An L-layer μP network learns a kernel λ_k^* and, against it, f^* has a *realised* source exponent r. Keeping t and r distinct is the move that untangles all subsequent arguments.

2. Why stationary learning has a ceiling

Start with what any estimator can do on a Sobolev ball. Packing $M \asymp \delta^{-d^*/s}$ perturbations at the resolution scale, each of $\mathcal{H}^s$-norm $\leq R$ and pairwise L^2-separated by $\asymp \varepsilon$, Fano's inequality forces error $\gtrsim \varepsilon^2$ whenever $D\varepsilon^2 \lesssim \log M$. Optimising over δ pins the rate to

$$D^{-\beta_0}, \qquad \beta_0 = \frac{2s}{2s + d^*}.$$

Feature learning is a special case of "estimator from D samples," so this bound applies to it unconditionally—the stationary attractor cannot escape it no matter how the kernel is tuned (**R1**).

Now ask what the attractor actually does. The μP mean-field closure reduces kernel dynamics to a spectral gradient system with effective potential $V(\boldsymbol{\lambda}) = \sum_k \bar{a}_k^2/\lambda_k + \mu \sum_k \lambda_k^\nu$, where the capacity term $\sum_k \lambda_k^\nu$ is derived, not postulated: for a depth-L diagonal network the rich-regime implicit bias selects $\nu = 1/L$ by the Schatten argument of Gunasekar et al. and Arora et al. (**R3**, unconditional for deep-linear, conditional on eigenbasis stability for general μP).

Setting $\partial V/\partial \lambda_k = 0$ gives the attractor $\lambda_k^* \propto (\bar{a}_k^2)^{1/(\nu+1)} \asymp k^{-b}$ with $b = (1 + 2t)/(\nu + 1)$. The realised source exponent of f^* against this kernel works out to

$$r(\nu) = \frac{t(\nu + 1)}{1 + 2t},$$

strictly increasing in ν, reaching $r = \frac{1}{2}$ at $\nu = 1/(2t)$ and nowhere above it under the barrier. So the attractor self-tunes toward $r = \frac{1}{2}$—and the minimax bound says it cannot go further. Stationary feature learning self-organises to the boundary of $\mathcal{H}^s$ and stays there (**R2**).

Two parameters enter separately: t is a property of the data; $\nu = 1/L$ is a property of the architecture. The value $r = \frac{1}{2}$ is their coincidence, not an identity built into the setup.

3. Why adaptivity does not help on Sobolev data

One might hope that adaptively choosing *which* features to use—resolving only the most informative modes—produces a better approximation exponent. Classical nonlinear approximation theory says no: for the Sobolev ball $\mathcal{H}^s(\mathcal{M})$ the nonlinear (best N-term) width matches the linear width,

$$d_N(\mathcal{H}^s) \asymp \delta_N(\mathcal{H}^s) \asymp N^{-s/d^*},$$

so $\alpha = 2s/d^*$ holds regardless of how cleverly the features are chosen, and $\alpha > \beta_0$ always (**R4**). The Laplacian eigenbasis is already rate-optimal for $\mathcal{H}^s$; sorting buys nothing.

A genuine exponent gain therefore requires the target to have *less* effective dimension than d^*. This happens in exactly two ways: **(a)** a compositional / hierarchical target $f^* = g_L \circ \cdots \circ g_1$ with each piece of local dimension d_i, giving $\alpha = 2s/d^*_{\text{loc}}$ where $d^*_{\text{loc}} = \max_i d_i < d^*$; or **(b)** transient, non-stationary kernel alignment. Any empirically observed $\beta > \beta_0$ or accelerated α must come from one of these two exits—the stationary attractor contributes neither.

4. The phase transition and when emergence is real

With both attractors characterised, the phase transition follows directly. A network resolving the compositional flag needs enough parameters to express it: the approximation cost of hitting local dimension d^*_{loc} to accuracy ε scales as $\varepsilon^{-d^*_{\text{loc}}/s}$, and the factorisation chain stays open only when each level's interaction gap $\Delta_i = \sigma_{\min}(\mathbb{E}[\nabla^2 g_i]) > 0$. These two requirements together define a finite capacity threshold $N_c \asymp \Delta_{\min}^{-d^*_{\text{loc}}/s}$, independent of the sample budget D.

Below N_c the network cannot instantiate the compositional flag with a positive gap—it sits on the stationary attractor with exponents (α, β_0). Above N_c it snaps onto the compositional attractor with exponents $(\alpha', \beta') = (2s/d^*_{\text{loc}},\ 2s/(2s + d^*_{\text{loc}}))$. The jump in the data exponent at N_c is

$$\Delta_\beta = \frac{2s(d^* - d^*_{\text{loc}})}{(2s + d^*_{\text{loc}})(2s + d^*)},$$

and this leads directly to the criterion the Schaeffer–Wei debate lacked:

$$\boxed{\Delta_\beta > 0 \iff d^*_{\text{loc}} < d^*.}$$

When $d^*_{\text{loc}} = d^*$ the defect $\mathfrak{d}(\phi, \mathcal{F})$ is smooth in N along the single frontier N^{-2s/d^*}; any apparent jump under a nonlinear metric is a metric artifact, exactly Schaeffer's mirage. When $d^*_{\text{loc}} < d^*$ the subspace V_ϕ snaps from a full-d^* block onto the d^*_{loc}-flag—a genuine L^2 discontinuity, exactly Wei's real jump. Both camps are right, on disjoint subsets of tasks.

5. The fourth theory: abstractions

The third question—which structures let knowledge transfer—has a clean answer once the right object is identified. An *abstraction* $(\mathcal{Z}, \phi)$ has defect $\mathfrak{d}(\phi, \mathcal{F}) = \sup_{f \in \mathcal{F}} \inf_g \|f - g \circ \phi\|_{L^2}$. The key observation is that this defect sees ϕ only through the closed subspace $V_\phi = L^2(\sigma(\phi))$, with the optimal decoder given by conditional expectation $\mathbb{E}[f \mid \phi]$. From this projection identity four structural facts follow:

Compression is a nonlinear width: the compression–defect frontier decays at rate $m^{-2s/d^*_{\text{loc}}}$, recovering d^*_{loc} as the exponent of the concept class.

Transfer is bounded by subspace geometry: for two concept classes, $\mathfrak{d}(\phi_{\mathcal{F}_1}, \mathcal{F}_2) \le d_n(\mathcal{F}_2) + R\,\Theta(V_1, V_2)$, where Θ is the operator gap between the width-optimal subspaces.

Gluing local representations into a global one is obstructed by a Čech cohomology class: $\delta_{\text{glob}} \le \varepsilon_{\text{loc}} + C\,R(\mathcal{F})\,\|[g]\|_{\check{H}^1}$. The obstruction vanishes iff the transition cocycle is a coboundary.

Learnability is gated by the interaction gap $\Delta_2 = \sigma_{\min}(\mathbb{E}[\nabla^2 g])$. For the Gaussian multi-index model, Δ_2 equals the link function's Hessian floor; link ellipticity forces $\Delta_2 \ge \rho > 0$, and the statistical-query barrier is recovered at $\Delta_2 = 0$. This derives, rather than assumes, the threshold that earlier work treated as a condition.

6. What the benchmark shows

A dependency-free implementation (pure NumPy, flat torus $\mathbb{T}^2$) plants every theoretical quantity exactly and reads it back. All five predictions pass; every negative control fails as predicted:

- barrier confirmed at population level, $\beta_0 = 0.556$ inside the 90% CI $[0.555, 0.598]$;
- compositional approximation exponent jumps to $2s/d^*_{\text{loc}} = 2.5$, while the linear (Sobolev) curve reverts to $2s/d^* = 1.25$;
- attractor family $r(\nu)$ reproduced from random initialisation, crossing $r = \frac{1}{2}$ at $\nu^\star = 0.80$;
- trained-NTK source exponent $\hat{r} \approx 0.45$, flat across widths—the stationarity signature certifying $\beta = \beta_0$;
- lazy network and under-smoothing kernel fail decisively (85× test error).

7. What remains open

Two things are open, stated without ceremony.

Theoretically: the nonlinear-μP extension of R3—proving that the Schatten penalty survives beyond the diagonal/deep-linear case—reduces to a single named hypothesis about eigenbasis stability. In the weak-coupling phase this hypothesis is discharged via a DMFT closure with a tight no-free-lunch frontier; the strong-coupling phase remains the sole residue.

Experimentally: the decisive test is not performed here. It requires an LLM-scale training run with independently measured $(d^*, d^*_{\text{loc}}, s)$, tracking the source exponent $\hat{r}(t)$ shoulder by shoulder across the knee D_{cross}. The theory predicts flat $\frac{1}{2}$-plateaus punctuated by rising spikes at each compositional shoulder. Exhibiting this pattern at synthetic or small scale is not evidence for it—the discriminating content lives only on natural data past D_{cross}. The benchmark validates the instrumentation; the scale run is the open experiment.

The six manuscripts: Boundaries of Stationary Feature Learning (theory & barrier), Composable Abstractions (the fourth theory), Spectral Scaling (empirical benchmark), Phase Transition Theorem (synthesis & authenticity criterion), Absorption–Constraint Duality (non-asymptotic geometry), and this précis. Full PDFs and runnable code at the repository above.